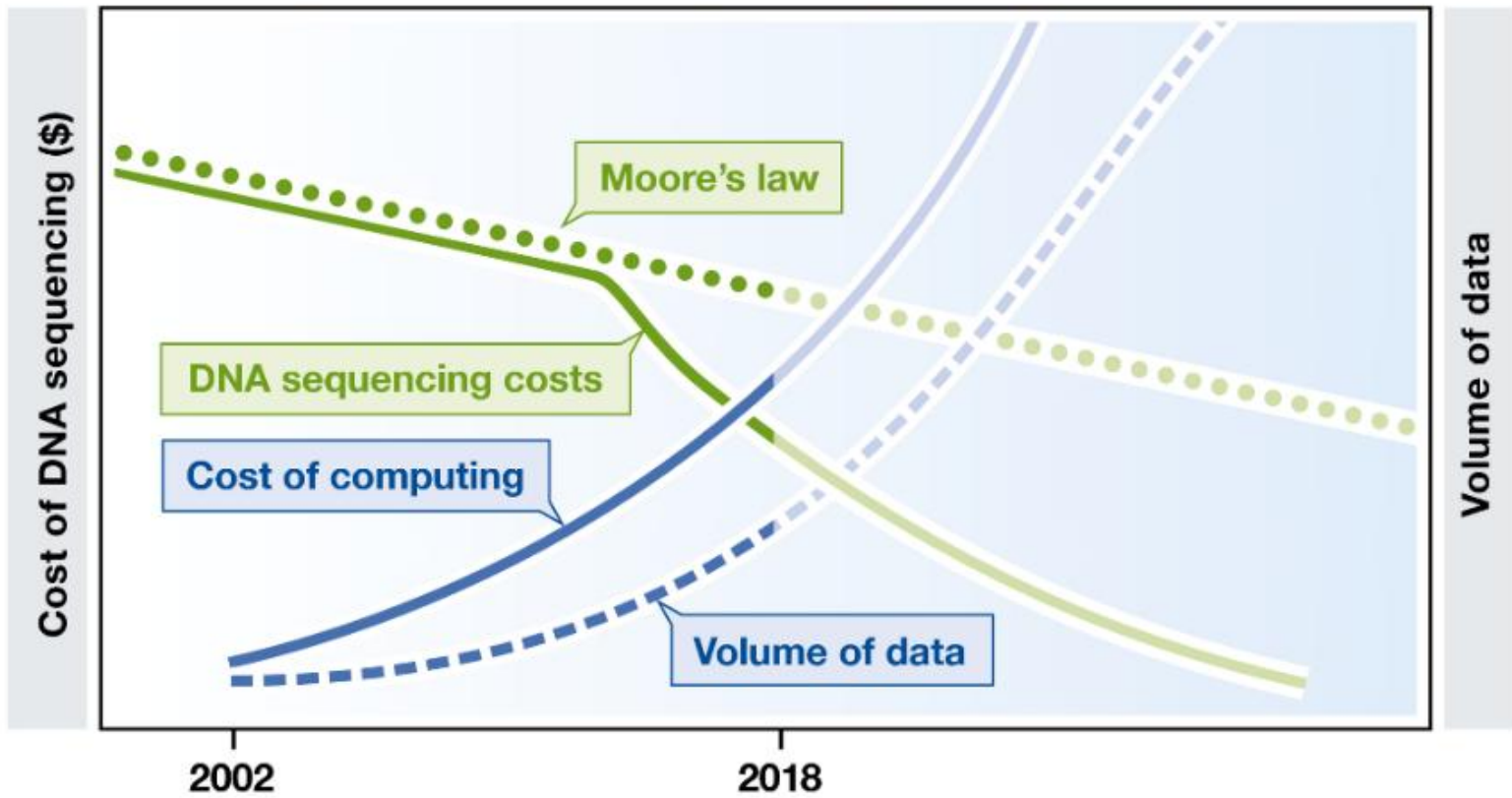
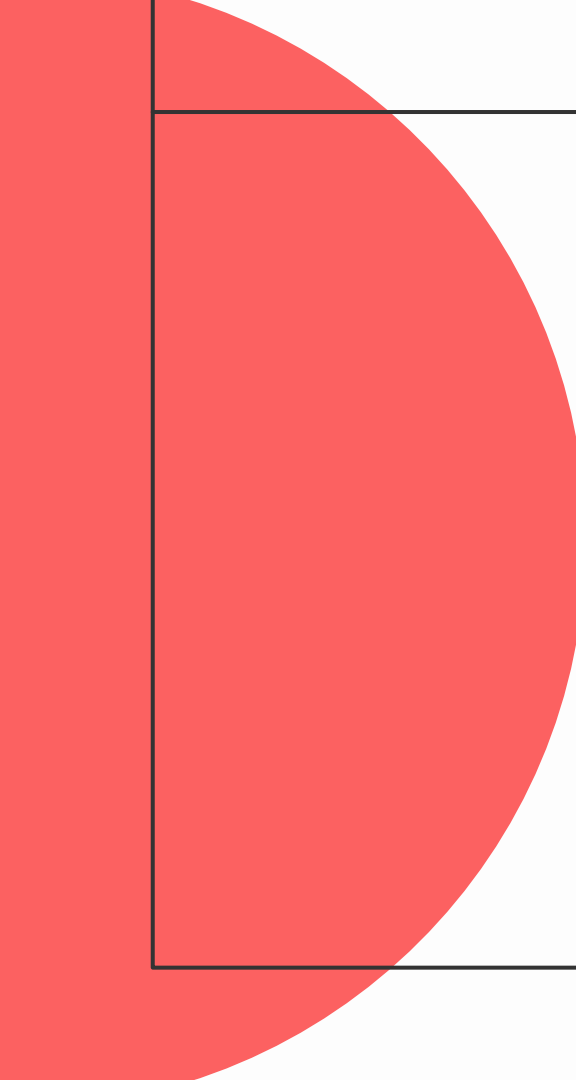




KUZMAN CONSULTING

BIOINFORMATICIANS AT YOUR SERVICE



A large red circle is positioned on the left side of the slide, partially cut off by the edge. It overlaps with a white rectangular area that contains the text.

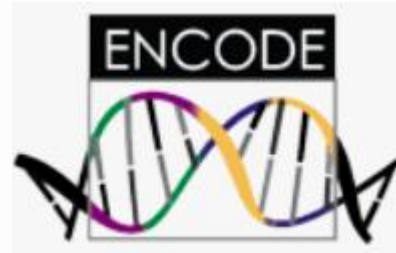
NGS baze podataka

Baze bioloških podataka

- podatci u različitom stupnju obrade
- ručno održavane ili “otvorene”



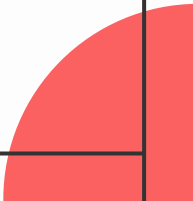
TACACAAATCAGTTAGTTT
GTCGGCAATCCGTAAGAT
AGCTTACACAACTGCAGAT
TAAAT
ATTTCGCGGCAAGCG
AATAATAAAACAACAAAC
AACTTCTGCCTGCACCTTG



NCBI

- skup međusobno povezanih baza podataka
- dobra početna točka
- sekvence, anotacije, literature, varijante, taksonomija...
- dobra dokumentacija
- napredno pretraživanje

<https://www.ncbi.nlm.nih.gov/>

A red circular graphic element is located in the bottom right corner of the slide, partially cut off by the edge.

PubMed

- medicinska i biološka literatura

PubMed Advanced Search Builder

 [User Guide](#)

Add terms to the query box

Author

Zhou

×

ADD

▼

Add with AND

Add with OR

Add with NOT

Query box

Enter / edit your search query here

Nucleotide

- [GenBank](#)
- [RefSeq](#)
- [SRA](#)
- [Gene](#)
- [Genome](#)

Nucleotide

Nucleotide

BRCA2 AND srcdb_refseq[PROP]

Create alert Advanced

Help

Species

Animals (21,186)

Plants (188)

Fungi (526)

Protists (221)

Bacteria (6)

Viruses (19)

Customize ...

Molecule types

genomic DNA/RNA (9,407)

mRNA (13,231)

Customize ...

Source databases

INSDC (GenBank) (6,207)

RefSeq (17,068)

Customize ...

Sequence Type

Nucleotide (23,131)

EST (201)

GSS (2)

Genetic compartments

Mitochondrion (1)

Sequence length

Custom range...

Release date

Custom range...

Revision date

Custom range...

Summary

20 per page

Sort by Default order

Send to:

Filters: [Manage Filters](#)

Items: 1 to 20 of 17068

<< First < Prev Page 1 of 854 Next > Last >>

☐ [Trypanosoma melophagium BRCA2 \(LSM04_005978\). partial mRNA](#)

1. 3,117 bp linear mRNA

Accession: XM_067337953.1 GI: 2787691070

[BioProject](#) [BioSample](#) [Protein](#) [Taxonomy](#)

[GenBank](#) [FASTA](#) [Graphics](#)

☐ [Trypanosoma equiperdum BRCA2 \(TEOVI_000401200\). partial mRNA](#)

2. 2,907 bp linear mRNA

Accession: XM_067226831.1 GI: 2785790759

[BioProject](#) [BioSample](#) [Protein](#) [Taxonomy](#)

[GenBank](#) [FASTA](#) [Graphics](#)

☐ [Trypanosoma melophagium isolate St. Kilda chromosome Unknown.ctg10segment3. whole genome shotgun sequence](#)

3. 396,957 bp linear DNA

Accession: NW_027125136.1 GI: 2787724863

[Assembly](#) [BioProject](#) [BioSample](#) [Protein](#) [Taxonomy](#)

[GenBank](#) [FASTA](#) [Graphics](#)

Nucleotide

Nucleotide

BRCA2

Create alert Advanced

Help

Summary

20 per page

Sort by Default order

Send to:

Filters: [Manage Filters](#)

GENE

Was this helpful?

[BRCA2 -- BRCA2 DNA repair associated](#)

[Homo sapiens \(human\)](#)

Also known as: BRCC2, BROVCA2, FACD, FAD, FAD1, FANCD, FANCD1, GLM3, PNCA2, XRCC11

Gene ID: 675

[RefSeq products](#) [Orthologs](#) [Genome Data Viewer](#)

[New - Visualize gene across multiple species](#)

RefSeq Sequences

+

Items: 1 to 20 of 23334

<< First < Prev Page 1 of 1167 Next > Last >>

☐ [Homo sapiens isolate B47 BRCA2 \(BRCA2\) gene. partial cds](#)

1. 287 bp linear DNA

Accession: MF096562.1 GI: 1388694389

[Protein](#) [Taxonomy](#)

[GenBank](#) [FASTA](#) [Graphics](#)

☐ [Homo sapiens isolate B54 BRCA2 \(BRCA2\) gene. partial cds](#)

2. 287 bp linear DNA

Accession: MF096561.1 GI: 1388694387

[Protein](#) [Taxonomy](#)

[GenBank](#) [FASTA](#) [Graphics](#)

☐ [Homo sapiens isolate B49 BRCA2 \(BRCA2\) gene. partial cds](#)

3. 287 bp linear DNA

Results by taxon

Top Organisms [Tree](#)

Homo sapiens (749)

Mus musculus (168)

Ovis aries (92)

Cricetus griseus (91)

Macaca fascicularis (77)

All other taxa (15891)

More...

Find related data

Database: [Select](#)

[Find Items](#)

Search details

BRCA2[All Fields] AND srcdb_refseq[PROP]

Top Organisms [Tree](#)

Homo sapiens (1841)

unidentified (614)

Mus musculus (440)

synthetic construct (140)

Canis lupus familiaris (140)

All other taxa (20153)

More...

Find related data

Database: [Select](#)

[Find Items](#)

Search details

BRCA2[All Fields]

Recent activity

[Turn Off](#) [Clear](#)

[Q](#) BRCA2 (23334) [Nucleotide](#)

[Q](#) Profile neighbors for GEO Profiles (Select 78694385) (199) [GEO Profiles](#)

[Q](#) CYP11A1 cytochrome P450 family 1 subfamily A member 1 [Homo sapiens] [Gene](#)

[Q](#) Gene Links for GEO Profiles (Select 78694385) (1) [Gene](#)

[Q](#) Mutant p53 depletion effect on MDA-468-shp53 breast tumor cell line [GEO Datasets](#)

Protein

- CDD
- Structure

Citation: ?

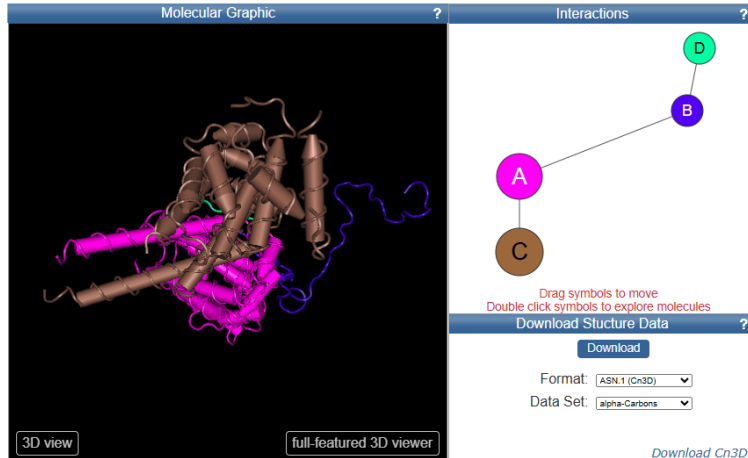
Structure of a meiosis-specific complex central to BRCA2 localization at recombination sites

Pendlebury DF, Zhang J, Agrawal R, Shibuya H, Nandakumar J
Nat Struct Mol Biol (2021) **28** p.671-680

Abstract

Meiotic cells invoke breast cancer susceptibility gene 2 (BRCA2) to repair programmed double-stranded DNA breaks and accomplish homologous recombination. The meiosis-specific protein MEILB2 facilitates BRCA2 recruitment to meiotic recombination sites. Here, we combine crystallography, biochemical analysis and a mouse meiosis model to reveal a robust architecture that ensures...
[read more](#)

PDB ID: 7LDG [Download](#) ?
MMDB ID: 205498 ?
PDB Deposition Date: 2021/1/13 ?
Updated in MMDB: 2024/10 ?
Experimental Method: x-ray diffraction ?
Resolution: 2.56 Å ?
Source Organism: Homo sapiens ?
Similar Structures: [VAST+](#) ?
[Download sequence data](#) ?



pfam00634: **BRCA2** (this model, PSSM-Id:459881 is obsolete and has been replaced by 459881)

[Download alignment](#) ?

BRCA2 repeat

The alignment covers only the most conserved region of the repeat.

Links ?

Source: pfam
Taxonomy: Eukaryota
PubMed: 1 link
Protein: Representatives
Specific Protein
Related Protein
Related Structure
Architectures
Superfamily: c02912

Statistics ?

Structure ?

PubMed References ?

Internal repeats in the BRCA2 protein sequence. *Nat Genet* 1996 May ; 13(1):22-3

Sequence trees and CD hierarchy information associated with this CD can not be displayed.

This model has been replaced with **513619**.

Sequence Alignment

☒ include consensus sequence ?

Format: [Hypertext](#) ▼ Row Display: [up to 10](#) ▼ Color Bits: [2.0 bit](#) ▼ Type Selection: [the most diverse members](#) ▼

XP_001237134	909	FGGGFSTAGGGKTSVSKEALERAQKAF	937
XP_010727291	2023	PALGFSTASGKQTISEAYRKAKALLKE	2051
ERH95536	176	GAPLFRSGHGKPGVAESSIRKAMAVLGE	204
EGV98880	1637	SALCTDTGHRKTCVSESSLKDRKWLRLD	1665
P97929	1626	SALAYTTEDSRKTCVRESLSKGRKWLRL	1654
XP_019695293	1668	SALSFYTGHRKTSVSQSLSLEVKWLRL	1696
XP_023403030	1650	TALAFYTGGRQTSVSQTSLSLEAKWVRE	1678
XP_012665452	1667	SALAFYTGHRKTSVSQASLFAKKWLRL	1695
ELK09316	1682	SVLAFYTGHRKTSVRETSLSLEARKWLRL	1710
XP_012039996	2058	AFSGFSTASGKHVPVESALRKVKGMFEE	2086

Citing CDD

Jiyao W et al.(2023). "The conserved domain database in 2023.", *Nucleic Acids Res.* **51**(D1):D384-D388.

Variation

- dbSNP
- ClinVar

ClinVar

[Create alert](#)
[Advanced](#)

[Home](#)
[About](#)
[Access](#)
[Help](#)
[Submit](#)
[Statistics](#)
[FTP](#)


 We've updated the full submission template; previous versions are no longer supported. Download the latest version from [the ClinVar FTP site](#) and [read more about the available templates](#).

Classification type

☐ Germline (21,055)

☐ Somatic (9)

Germline classification

☐ Conflicting classifications (5,629)

☐ Benign (902)

☐ Likely benign (4,802)

☐ Uncertain significance (4,033)

☐ Likely pathogenic (725)

☐ Pathogenic (5,393)

Types of conflicts

☐ P/LP vs L/B/B (7)

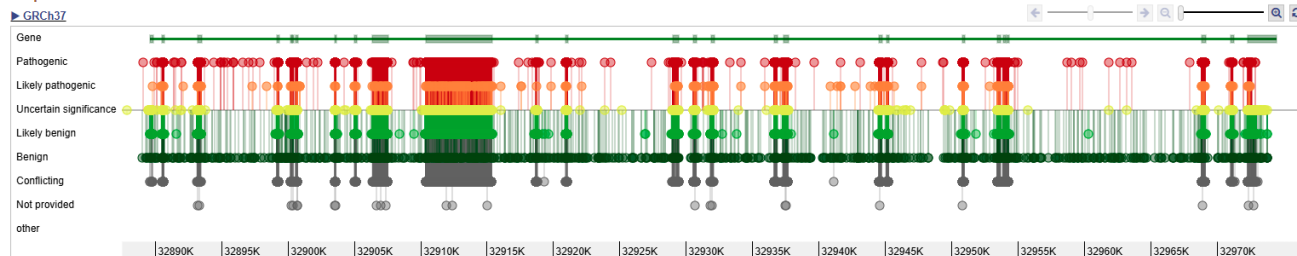
☐ P/LP vs VUS (111)

☐ VUS vs L/B/B (5,521)

Molecular consequence

☐ Frameshift (3,679)

Graphical view of search results ▲



GEO (Gene Expression Omnibus)

- ekspresija
- microarray,
RNAseq, rtPCR,
SAGE...

 **National Library of Medicine**
National Center for Biotechnology Information

Log in

GEO Profiles

Advanced [Help](#)



GEO Profiles

This database stores individual gene expression profiles from curated DataSets in the Gene Expression Omnibus (GEO) repository. Search for specific profiles of interest based on gene annotation or pre-computed profile characteristics.

Getting Started

- [GEO Documentation](#)
- [GEO FAQ](#)
- [About GEO Profiles](#)
- [Construct a Query](#)
- [Download Options](#)

GEO Tools

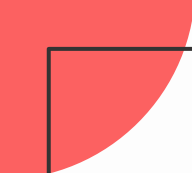
- [Submit to GEO](#)
- [Advanced Search](#)
- [DataSet Browser](#)
- [Programmatic Access](#)

More Resources

- [GEO Home](#)
- [GEO DataSets](#)
- [SRA](#)

Example Searches

Gene symbol	CYP1A1[Gene Symbol]
Gene symbols in DataSets that contain specific keywords	(CYP1A1[Gene Symbol] OR ME1[Gene Symbol]) AND (smok* OR diet)
Partial gene name in a specific DataSet	kinase[Gene Description] AND GDS182
Gene Ontology(GO) term in a specific DataSet	apoptosis[Gene Ontology] AND GDS182
Chromosome region and species	(8[Chromosome] AND 10000:3000000[Base Position]) AND mouse[organism]
Genes that show subset effects in DataSets that examine the effect of an agent	agent[Flag Information] AND "value subset effect"[Flag Type]



Nucleotide

- GenBank
- RefSeq
- SRA

Protein

- CDD
- Structure

Variation

- dbSNP
- ClinVar

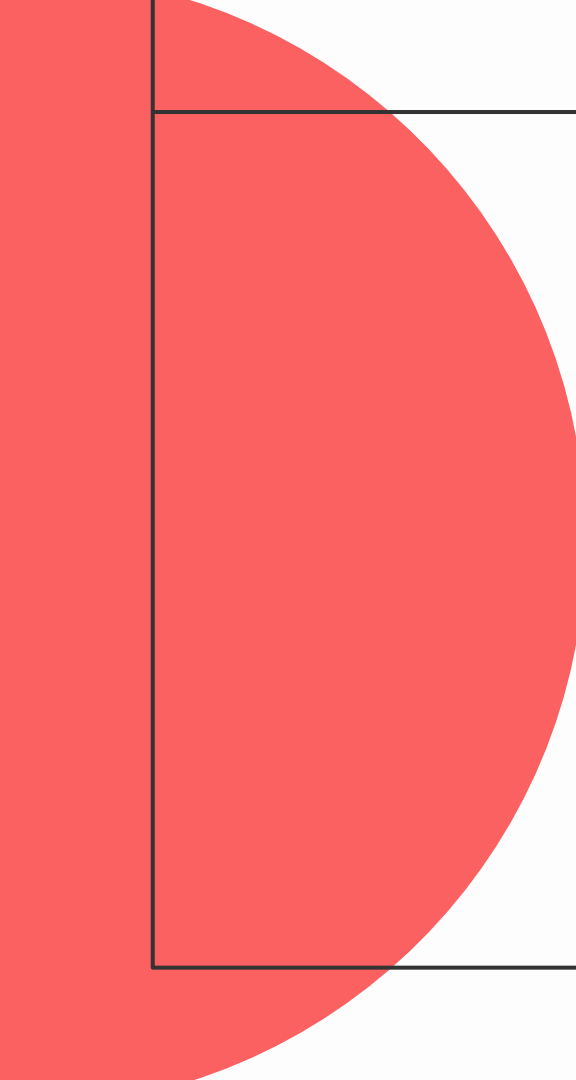
Genes & Expression

- BioProject
- GEO
- Online Mendelian Inheritance in Man (OMIM)

Homology

Taxonomy

Data & Software

A large red circular graphic element is positioned on the left side of the slide, partially overlapping the white content area.

Genomski preglednici

Ensembl

- provjereni podatci
- anotacija
- vizualizacija!

 [BLAST/BLAT](#) | [VEP](#) | [Tools](#) | [BioMart](#) | [Downloads](#) | [Help & Docs](#) | [Blog](#)

[Tools](#)
[All tools](#)

[BioMart >](#)
Export custom datasets from Ensembl with this data-mining tool

[BLAST/BLAT >](#)
Search our genomes for your DNA or protein sequence

[Variant Effect Predictor >](#)
Analyse your own variants and predict the functional consequences of known and unknown variants

Search

Human

for

BRCA2

Go

e.g. [BRCA2](#) or [rat 5:62797383-63627669](#) or [rs699](#) or [coronary heart disease](#)

All genomes

-- Select a species --

 **Pig breeds**
Pig reference genome and 20 additional breeds

[View full list of all species](#)

Favourite genomes 

 **Human**
GRCh38.p14
[Still using GRCh37?](#)

 **Mouse**
GRCm39

 **Zebrafish**
GRCz11



Human (GRCh38.p14) ▼

Location: 13:32,313,383-32,401,971

Gene: BRCA2

Location-based displays

- Whole genome
- Chromosome summary
- Region overview
- Region in detail**
- Comparative Genomics
 - Synteny
 - Alignments (image)
 - Alignments (text)
 - Region Comparison
- Genetic Variation
 - Variant table
 - Resequencing
 - Linkage Data
- Markers
- Other genome browsers
 - UCSC
 - NCBI
 - Ensembl GRCh37

Configure this page

Custom tracks

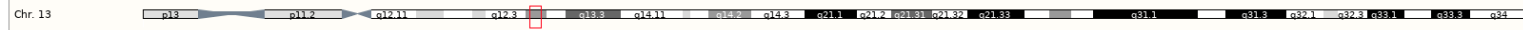
Export data

Share this page

Bookmark this page

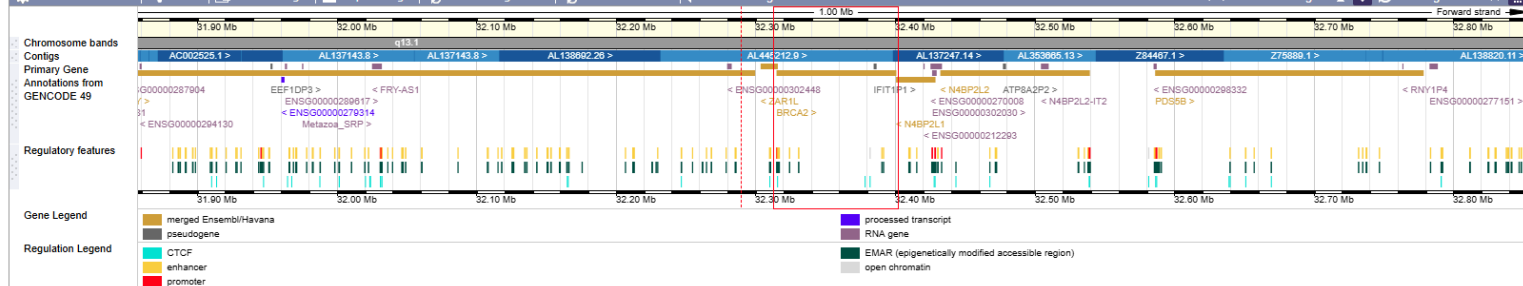
Chromosome 13: 32,313,383-32,401,971

Add/remove tracks | Share | Export image | Reset configuration



Region in detail

Add/remove tracks | Share | Resize image | Export image | Reset configuration | Reset track order | Switch image



Location: 13:32313383-32401971

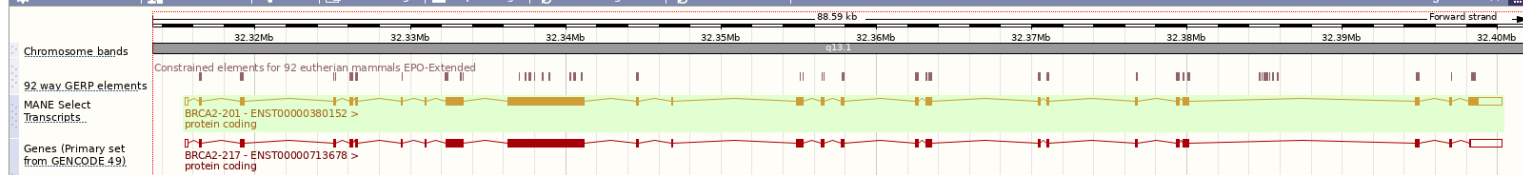
Go

Gene:

Go

Add/remove tracks | Custom tracks | Share | Resize image | Export image | Reset configuration | Reset track order

Drag/Select



UCSC Genome Browser

UNIVERSITY OF CALIFORNIA
SANTA CRUZ

Genomics
Institute



Genome Browser Gateway



Genomes

Genome Browser

Tools

Mirrors

Downloads

My Data

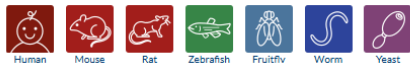
Projects

Help

About Us

Browse/Select Species

POPULAR SPECIES



Search through thousands of genome assemblies ①

Enter species, common name or assembly ID

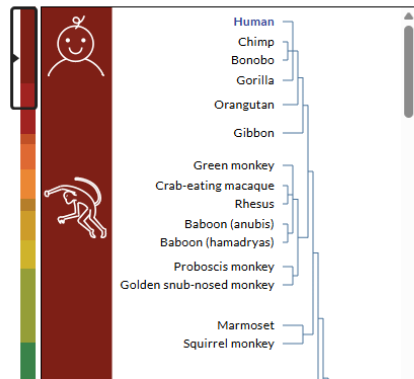
[Unable to find a genome? Send us a request.](#)

RECENT GENOMES ①

Search for a genome above, or click a popular species icon

[Hide species tree](#)

COMMONLY USED ASSEMBLIES



Find Position

Human Assembly

Feb. 2009 (GRCh37/hg19)



Position/Search Term (Examples)

Enter position, gene symbol or search terms

Current position: chr1:74,486,865-74,487,865

Human Genome Browser - hg19 assembly

[View sequences](#)

The February 2009 human reference sequence (GRCh37) was produced by the [Genome Reference Consortium](#). For more information about this assembly, see [GRCh37](#) in the NCBI Assembly database.

Sample position queries

A genome position can be specified by the accession number of a sequenced genomic clone, an mRNA or EST or STS marker, a chromosomal coordinate range, or keywords from the GenBank description of an mRNA. The following list shows examples of valid position queries for the human genome. See the [User's Guide](#) for more information.

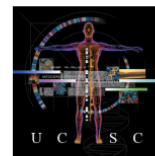
Request:

Genome Browser Response:

chr7	Displays all of chromosome 7
chrUn_gl000212	Displays all of the unplaced contig gl000212
20p13	Displays region for band p13 on chr 20
chr3:1-1000000	Displays first million bases of chr 3, counting from p-arm telomere
chr3:1000000+2000	Displays a region of chr3 that spans 2000 bases, starting with position 1000000

RH18061;RH80175	Displays region between genome landmarks, such as the STS markers RH18061 and RH80175, or chromosome bands 15q11 to 15q13, or SNPs rs1042522 and rs1800370. This syntax may also be used for other range queries, such as between uniquely determined ESTs, mRNAs, refSeqs, etc.
15q11:15q13	
rs1042522;rs1800370	

D16S3046	Displays region around STS marker D16S3046 from the Genethon/Marshfield maps. Includes 100,000 bases on each side as well.
AA205474	Displays region of EST with GenBank accession AA205474 in BRCA1 cancer gene on chr 17
AC008101	Displays region of clone with GenBank accession AC008101
AF083811	Displays region of mRNA with GenBank accession number AF083811
PRNP	Displays region of genome with HUGO Gene Nomenclature Committee Identifier PRNP
NM_017414	Displays the region of genome with RefSeq identifier NM_017414
NP_059110	Displays the region of genome with protein accession number NP_059110



Homo sapiens

(Graphic courtesy of CBSE)



brojne opcije, fleksibilno, mogućnost dodavanja vlastitih podataka!

Collapse all

Track search

Highlight

Hide all

Manage custom tracks

Configure

Reverse

Resize

Expand all

Visible Tracks

Hide group Refresh

My Track	GENCODE V49lift37	NCBI RefSeq	ClinVar Variants	1000G Archive	Updated gnomAD	CGAP SAGE	GTEx Gene V8
Dense ▾	Pack ▾	Dense ▾	Dense ▾	Show ▾	Show ▾	Full ▾	Full ▾

Custom Tracks

Hide group Refresh

My Track
Dense ▾

Mapping and Sequencing

Hide group Refresh

Base Position	Assembly	BAC End Pairs	BU ORCID	Chromosome Band	deCODE Recomb	ENCODE Pilot	Exome Probesets
Dense ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾
FISH Clones	Fosmid End Pairs	Gap	GC Percent	GRC Incident	GRC Map Contigs	GRC Patches	Hg18 Diff
Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾
Hg38 Diff	Hi Seq Depth	INSDC	liftOver & ReMap	LRG Regions	Map Contigs	Mappability	Problematic Regions
Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾
Recomb Rate	RefSeq Acc	Restr Enzymes	Short Match	STS Markers			
Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾			

Genes and Gene Predictions

Hide group Refresh

GENCODE V49lift37	NCBI RefSeq	Genes Archive	CCDS	CRISPR Targets	Ensembl Genes	EvoFold	Exoniphy
Pack ▾	Dense ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾
GENCODE Versions	H-Inv 7.0	HGNC	IKMC Genes Mapped	lincRNAs	LRG Transcripts	MGC/ORFeome Genes	Other RefSeq
Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾
Plam in GENCODE	Prediction Archive	Retroposed Genes	sno/miRNA	TransMap V5	tRNA Genes	UCSC Alt Events	UniProt
Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾
Vega Genes	Yale Pseudo60						
Hide ▾	Hide ▾						

Phenotypes, Variants, and Literature

Hide group Refresh

OMIM	Publications	AlphaMissense	CADD 1.6	CADD 1.7	CIViC	ClinGen	ClinGen CNVs
Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾
ClinVar Variants	Constraint scores	Coriell CNVs	COSMIC	COSMIC Regions	COVID Data	DECIPHER	Deleteriousness Predictions
Dense ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾
Development Delay	Dosage Sensitivity	G2P Project	GAD View	GenCC	Gene Interactions	GeneReviews	GWAS Catalog
Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾
Haploinsufficiency	HGMD Public 2025	Lens Patents	LOVD Variants	MGI Mouse QTL	MITOMAP	Orphanet	PanelApp
Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	[No data-chr13]	Hide ▾	Hide ▾
Polygenic Risk Scores	REVEL Scores	RGD Human QTL	RGD Rat QTL	SNPedia	Splicing Impact	UniProt Variants	Updated Variants in Papers
Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾	Hide ▾
Web Sequences							
Hide ▾							

[Home](#) [Genomes](#) [Genome Browser](#) [Tools](#) [Mirrors](#) [Downloads](#) [My Data](#) [Projects](#) [Help](#) [About Us](#)

Add Custom Tracks

Clade [Mammal](#) Genome [Human](#) Assembly [Dec. 2013 \(GRCh38/hg38\)](#)

Display your own data as custom annotation tracks in the browser. Data must be formatted in [bigBed](#), [bigBarChart](#), [bigChain](#), [bigGenePred](#), [bigInteract](#), [bigLolly](#), [bigMaf](#), [bigMethyl](#), [bigPsl](#), [bigWig](#), [BAM](#), [barChart](#), [VCF](#), [BED](#), [BED detail](#), [bedGraph](#), [bedMethyl](#), [broadPeak](#), [CRAM](#), [GFF](#), [GFF3](#), [GTF](#), [hic](#), [interact](#), [MAF](#), [narrowPeak](#), [Personal Genome SNP](#), [PSL](#), or [WIG](#) formats.

- You can paste just the URL to the file, without a "track" line, for bigBed, bigWig, bigGenePred, CRAM, BAM and VCF.
- To configure the display, set [track](#) and [browser](#) line attributes as described in the [User's Guide](#).

Examples are [here](#). If you do not have web-accessible data storage available, please see the [Hosting](#) section of the Track Hub Help documentation.

Please note a much more efficient way to load data is to use [Track Hubs](#), which are loaded from the [Track Hubs Portal](#) found in the menu under My Data.

Paste URLs or data:

Or upload: [Choose File](#) No file chosen [Submit](#)

[Clear](#)

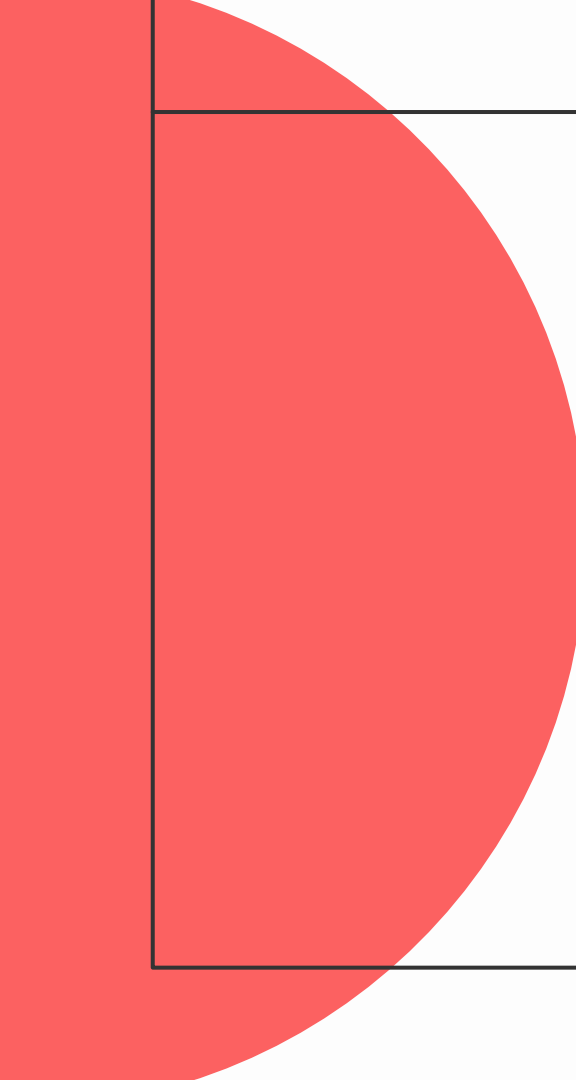
Optional track documentation:

Or upload: [Choose File](#) No file chosen

[Clear](#)

Click [here](#) for an HTML document template that may be used for Genome Browser track descriptions.

tutoriali

A large red semi-circle is positioned on the left side of the slide, partially overlapping a white rectangular area that contains the title.

Tipovi podataka

Genomski podatci

- biološki podatci su višedimenzionalni - izazov
- “sirove” sekvence, sekvence s meta-podatcima, sekvence u međusobnom odnosu, podatci o funkciji, jačina signala....
- plus meta-podatci o uzorku: priprema, histologija, lokacija, dob, spol....

Genomski podatci

- biološki podatci su višedimenzionalni - izazov
- “sirove” sekvence, sekvence s meta-podatcima, sekvence u međusobnom odnosu, podatci o funkciji, jačina signala....
- plus meta-podatci o uzorku: priprema, histologija, lokacija, dob, spol....
- neefikasno – većinom tekstualni formati
- nestandardizirano

HOW STANDARDS PROLIFERATE:
(SEE: A/C CHARGERS, CHARACTER ENCODINGS, INSTANT MESSAGING, ETC.)

SITUATION:
THERE ARE
14 COMPETING
STANDARDS.

14?! RIDICULOUS!
WE NEED TO DEVELOP
ONE UNIVERSAL STANDARD
THAT COVERS EVERYONE'S
USE CASES.



YEAH!

SOON:

SITUATION:
THERE ARE
15 COMPETING
STANDARDS.

FASTA i FASTQ

- osnovni format za spremanje (nukleotidnih) sekvenci

```
>sequenceID-001 description
```

```
AAGTAGGAATAATATCTTATCATTATAGATAAAAACCTTCTGAATTTGCTTAGTGTGTAT  
ACGACTAGACATATATCAGCTCGCCGATTATTTGGATTATTCCCTG
```

```
>sequenceID-002 description
```

```
CAGTAAAGAGTGGATGTAAGAACCGTCCGATCTACCAGATGTGATAGAGGTTGCCAGTAC  
AAAAATTGCATAATAATTGATTAATCCTTTAATATTGTTTAGAATATATCCGTCAGATAA  
TCCTAAAAATAACGATATGATGGCGGAAATCGTC
```

```
>sequenceID-003 description
```

```
CTTCAATTACCCTGCTGACGCGAGATACCTTATGCATCGAAGGTAAAGCGATGAATTTAT  
CCAAGGTTTTAATTTG
```

FASTA i FASTQ

- osnovni format za spremanje (nukleotidnih) sekvenci
- FASTQ sadrži i podatke o kvaliteti očitavanja

```
@A0216:173:HJNFKDSX:3:1101:2745:1016 1:N:0:TTACGCAC+AGATGGTC
GGGAGACGAGCGTCACGTTTCATGAGCGCCTCCTCGACCTCGTCGACGGAGTCGAAATTACGCGCCACCCACAGCGGAAATTCGAACGTGGCGACGTTTTCC
+
BBBFBFFFFFFFFBF<FBB0B<BF<FFFFFFIII'<<FF7BF<'7<BBBBB<<BBBBBBBBBBBBBBBBBBBBBB<7BB<<B<BFB<BBBB77<BB<BBBBB
@A0216:173:HJNFKDSX:3:1101:3884:1016 1:N:0:TTACGCAC+AGATGGTC
GCTTCGCCTTGCCGCCCCGACCAAACACCGTGTCATAGTGGTAGCCGCGCGGAGTAACCAAGATGGTCTCCCCATATGAGAACTCCAGTCGAGATTACGG
+
BBBBFFFFFFFFFFFFFFFFIBFFIIFFIIBBF0BFFF0BF<BF<BBFFBFFB<BBFBBB0<BBFFFBBBFBFFFBFFB<<BBBBFFFFB<BBFFB BBB07
@A0216:173:HJNFKDSX:3:1101:4010:1016 1:N:0:TTACGCAC+AGATGGTC
TATTTTATGCATTGCTTAATTTTAATAGCTGCAAAAGTTAATAAAATAAAACCAATTGCTACAAAAAAGAAGTGCAGCTTCAAGAAGATGTAAACTG
+
<<<<<0<00000<BB7BBBB000BB0<7BB000B<0BB'<<<'0'0'<0<00'07BB'77' '07<7' '0000'7<<0'0<<'7<'7'77'000'77' '7
```


FASTA i FASTQ

```
(33): !"#$%&'()*+,-./0123456789:;<=>?@ABCDEFGHI  
(64): @ABCDEFGHIJKLMNPOQRSTUVWXYZ[\]^_`abcdefgh
```

$$Q = -10 \log_{10} P \quad \longrightarrow \quad P = 10^{\frac{-Q}{10}}$$

Phred Quality Score	Probability of incorrect base call	Base call accuracy
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1000	99.9%
40	1 in 10000	99.99%
50	1 in 100000	99.999%

SAM i BAM

- spremanje mapiranih podataka
- zaglavlje + 11 obaveznih kolona

Col	Field	Type	Regex/Range	Brief description
1	QNAME	String	[!-?A-~]{1,254}	Query template NAME
2	FLAG	Int	$[0, 2^{16} - 1]$	bitwise FLAG
3	RNAME	String	* [:rname:^*=] [:rname:]*	Reference sequence NAME ¹²
4	POS	Int	$[0, 2^{31} - 1]$	1-based leftmost mapping POSition
5	MAPQ	Int	$[0, 2^8 - 1]$	MAPping Quality
6	CIGAR	String	* ([0-9]+[MIDNSHP=X])+	CIGAR string
7	RNEXT	String	* = [:rname:^*=] [:rname:]*	Reference name of the mate/next read
8	PNEXT	Int	$[0, 2^{31} - 1]$	Position of the mate/next read
9	TLEN	Int	$[-2^{31} + 1, 2^{31} - 1]$	observed Template LENgth
10	SEQ	String	* [A-Za-z=.]+	segment SEQUENCE
11	QUAL	String	[!-~]+	ASCII of Phred-scaled base QUALity+33

SAM i BAM

- spremanje mapiranih podataka
- zaglavlje + 11 obaveznih kolona – CIGAR string opisuje poravnanje

```
@HD VN:1.6 SO:coordinate
@SQ SN:ref LN:45
r001 99 ref 7 30 8M2I4M1D3M = 37 39 TTAGATAAAGGATACTG *
r002 0 ref 9 30 3S6M1P1I4M * 0 0 AAAAGATAAGGATA *
r003 0 ref 9 30 5S6M * 0 0 GCCTAAGCTAA * SA:Z:ref,29,-,6H5M,17,0;
r004 0 ref 16 30 6M14N5M * 0 0 ATAGCTTCAGC *
r003 2064 ref 29 17 6H5M * 0 0 TAGGC * SA:Z:ref,9,+,5S6M,30,1;
r001 147 ref 37 30 9M = 7 -39 CAGCGGCAT * NM:i:1
```

SAM i BAM

Header section

```
@HD    VN:1.3    SO:coordinate
@SQ    SN:contigA    LN:443
@SQ    SN:contigB    LN:1493
@SQ    SN:contigC    LN:328
```

Tab-delimited read alignment information lines

```
readID43GYAX15:7:1:1202:19894/1    256    contig43    613960    1    65M    *    0    0
CCAGCGCGAACGAAATCCGCATGCGTCTGGTCTGTGACGGAACGGCGGCGGTGTGATGCACGGC    EDDEEDEE=EE?DE??DDDBADEBEFFFD BEFFEBBCBC=?BEEEE@=:?:?:7?:8-
6?7?@??#    AS:i:0    XS:i:0    XN:i:0    XM:i:0    XO:i:0    XG:i:0    NM:i:0    MD:Z:65    YT:Z:UU
readID43GYAX15:7:1:1202:19894/1    272    contig32    21001    1    65M    *    0    0
GCCGGACGTACACGGCCGCCGGCGGCTCTACGACCAGACGCATGCGGATTCGTTAGAGCCGG    #??@?7?6-8:???:?:=@EEEEB?=CBCBEFFEBDFFE BEDABDDD??ED?
EE=EEDDEE    AS:i:-5    XS:i:0    XN:i:0    XM:i:1    XO:i:0    XG:i:0    NM:i:1    MD:Z:42T22    YT:Z:UU
readID43GYAX15:7:1:1202:19894/1    256    contig87    540849    1    65M    *    0    0
CCTGCACGAACGAAATCCGCATGCGTCTGGTCTGTGACGGAACGGCGTTGTGTGACGAACGGC    EDDEEDEE=EE?DE??DDDBADEBEFFFD BEFFEBBCBC=?BEEEE@=:?:?:7?:8-
6?7?@??# AS:i:0    XS:i:0    XN:i:0    XM:i:0    XO:i:0    XG:i:0    NM:i:0    MD:Z:65    YT:Z:UU
```

BED

- jednostavan tablični format

```
track name="AML v2 amplicons" description="Tapestri Designer: AML v2 panel: amplicon  
regions" type=bed db=hg19 color=77,175,74 priority=2  
chr1    115256487      115256723      AML_v2_NRAS_115256512  
chr1    115258609      115258825      AML_v2_NRAS_115258635  
chr2    25457143       25457372       AML_v2_DNMT3A_25457166  
chr2    25458518       25458763       AML_v2_DNMT3A_25458540  
chr2    25459793       25460046       AML_v2_DNMT3A_25459813  
chr2    25461880       25462137       AML_v2_DNMT3A_25461902  
chr2    25463106       25463346       AML_v2_DNMT3A_25463127  
chr2    25463493       25463717       AML_v2_DNMT3A_25463515  
chr2    25464421       25464618       AML_v2_DNMT3A_25464443  
chr2    25466621       25466871       AML_v2_DNMT3A_25466642  
chr2    25467012       25467220       AML_v2_DNMT3A_25467033  
chr2    25467371       25467631       AML_v2_DNMT3A_25467391  
chr2    25468110       25468332       AML_v2_DNMT3A_25468130  
chr2    25469006       25469231       AML_v2_DNMT3A_25469026
```

GTF GFF(3)

Gene Transfer
file Format

General
Feature Format

```
##gff-version 3.2.1
##sequence-region ctg123 1 1497228
ctg123 . gene 1000 9000 . + . ID=gene00001;Name=EDEN
ctg123 . TF_binding_site 1000 1012 . + . ID=tfbs00001;Parent=gene00001
ctg123 . mRNA 1050 9000 . + . ID=mRNA00001;Parent=gene00001;Name=EDEN.1
ctg123 . mRNA 1050 9000 . + . ID=mRNA00002;Parent=gene00001;Name=EDEN.2
ctg123 . mRNA 1300 9000 . + . ID=mRNA00003;Parent=gene00001;Name=EDEN.3
ctg123 . exon 1300 1500 . + . ID=exon00001;Parent=mRNA00003
ctg123 . exon 1050 1500 . + . ID=exon00002;Parent=mRNA00001,mRNA00002
ctg123 . exon 3000 3902 . + . ID=exon00003;Parent=mRNA00001,mRNA00003
ctg123 . exon 5000 5500 . + . ID=exon00004;Parent=mRNA00001,mRNA00002,mRNA00003
ctg123 . exon 7000 9000 . + . ID=exon00005;Parent=mRNA00001,mRNA00002,mRNA00003
ctg123 . CDS 1201 1500 . + 0 ID=cds00001;Parent=mRNA00001;Name=edenprotein.1
ctg123 . CDS 3000 3902 . + 0 ID=cds00001;Parent=mRNA00001;Name=edenprotein.1
ctg123 . CDS 5000 5500 . + 0 ID=cds00001;Parent=mRNA00001;Name=edenprotein.1
ctg123 . CDS 7000 7600 . + 0 ID=cds00001;Parent=mRNA00001;Name=edenprotein.1
ctg123 . CDS 1201 1500 . + 0 ID=cds00002;Parent=mRNA00002;Name=edenprotein.2
ctg123 . CDS 5000 5500 . + 0 ID=cds00002;Parent=mRNA00002;Name=edenprotein.2
ctg123 . CDS 7000 7600 . + 0 ID=cds00002;Parent=mRNA00002;Name=edenprotein.2
ctg123 . CDS 3301 3902 . + 0 ID=cds00003;Parent=mRNA00003;Name=edenprotein.3
ctg123 . CDS 5000 5500 . + 1 ID=cds00003;Parent=mRNA00003;Name=edenprotein.3
ctg123 . CDS 7000 7600 . + 1 ID=cds00003;Parent=mRNA00003;Name=edenprotein.3
ctg123 . CDS 3391 3902 . + 0 ID=cds00004;Parent=mRNA00003;Name=edenprotein.4
ctg123 . CDS 5000 5500 . + 1 ID=cds00004;Parent=mRNA00003;Name=edenprotein.4
ctg123 . CDS 7000 7600 . + 1 ID=cds00004;Parent=mRNA00003;Name=edenprotein.4
```

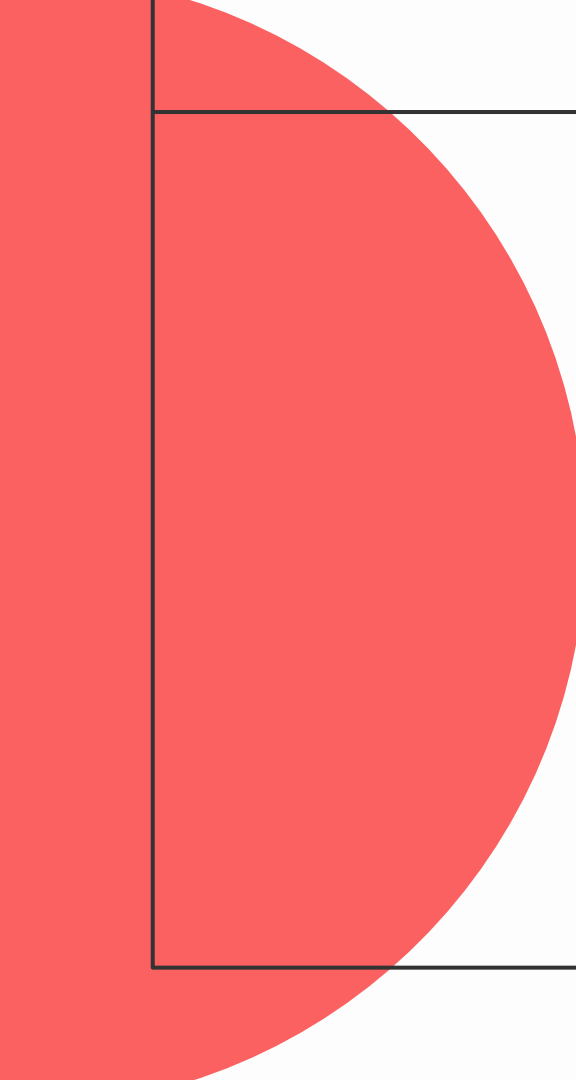

VCF

- Variant Call Format

```
##fileformat=VCFv4.3
##fileDate=20090805
##source=myImputationProgramV3.1
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=<ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FILTER=<ID=q10,Description="Quality below 10">
##FILTER=<ID=s50,Description="Less than 50% of samples have data">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype Quality">
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT NA000001 NA000002 NA000003
20 14370 rs6054257 G A 29 PASS NS=3;DP=14;AF=0.5;DB;H2 GT:GQ:DP:HQ 0|0:48:1:51,51 1|0:48:8:51,51 1/1:43:5:.,.
20 17330 . T A 3 q10 NS=3;DP=11;AF=0.017 GT:GQ:DP:HQ 0|0:49:3:58,50 0|1:3:5:65,3 0/0:41:3
20 1110696 rs6040355 A G,T 67 PASS NS=2;DP=10;AF=0.333,0.667;AA=T;DB GT:GQ:DP:HQ 1|2:21:6:23,27 2|1:2:0:18,2 2/2:35:4
20 1230237 . T . 47 PASS NS=3;DP=13;AA=T GT:GQ:DP:HQ 0|0:54:7:56,60 0|0:48:4:51,51 0/0:61:2
20 1234567 microsat1 GTC G,GTCT 50 PASS NS=3;DP=9;AA=G GT:GQ:DP 0/1:35:4 0/2:17:2 1/1:40:3
```

Specifikacije formata:

- <https://www.ensembl.org/info/website/upload/index.html>
- <https://genome.ucsc.edu/FAQ/FAQformat.html>
- <https://genome.ucsc.edu/ENCODE/fileFormats.html>

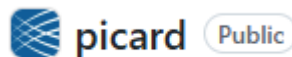
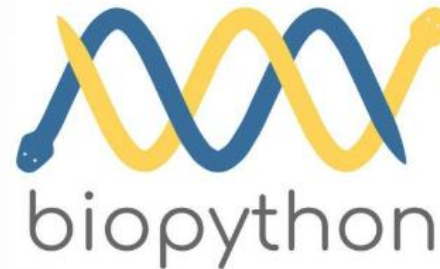
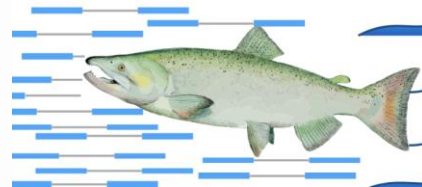
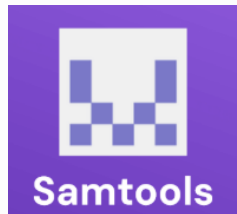
A large red circle is partially visible on the left side of the slide, overlapping the white content area.

Bioinformatički alati

Alati u bioinformatici

lh3/bwa

Burrow-Wheeler Aligner for short-read alignment
(see minimap2 for long-read alignment)



Alati u bioinformatiči - prednosti

- velik broj alata za skoro sve tipove analiza
- većina je open-source
- forumi i zajednice: SEQanswers, Biostars, GitHub issues



Alati u bioinformatiči - mane

- velik broj alata za skoro sve tipove analiza
- većina je open-source



WSL-Ubuntu

TerminalSessionsViewX serverToolsGamesSettingsMacrosHelp

SessionServersToolsGamesSessionsViewSplitMultiExecTunnelingPackagesSettingsHelp

X serverExit

Quick connect...

/home/dunja/

Name

..

.cache

.conda

.config

.cookiecutter_replay

.cpanm

.dotnet

.local

.m2

.nextflow

.nf-test

.nfcare

.npm

.sdkman

.ssh

.vscode-remote-containers

.vscode-server

Remote monitoring

☒ Follow terminal folder

2. WSL-Ubuntu

• MobaXterm Personal Edition v25.2 •
(SSH client, X server and network tools)

► Linux distribution: Ubuntu

► Windows drives are mounted into /mnt path (by default)

► WSL DISPLAY is automatically redirected to Windows desktop

► WSL filesystem is accessible in the sidebar browser

► For more info, ctrl+click on help or visit our website.

Welcome to Ubuntu 22.04.4 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

* Documentation: <https://help.ubuntu.com>

* Management: <https://landscape.canonical.com>

* Support: <https://ubuntu.com/pro>

This message is shown once a day. To disable it please create the /home/dunja/.hushlogin file.

-bash: export: `Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux_2.1.5.0_x64__8wekyb3d8bbwe:/mnt/c/Users/dunja/AppData/Roaming/MobaXterm/slash/bin:/mnt/c/WINDOWS:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/c/Program': not a valid identifier

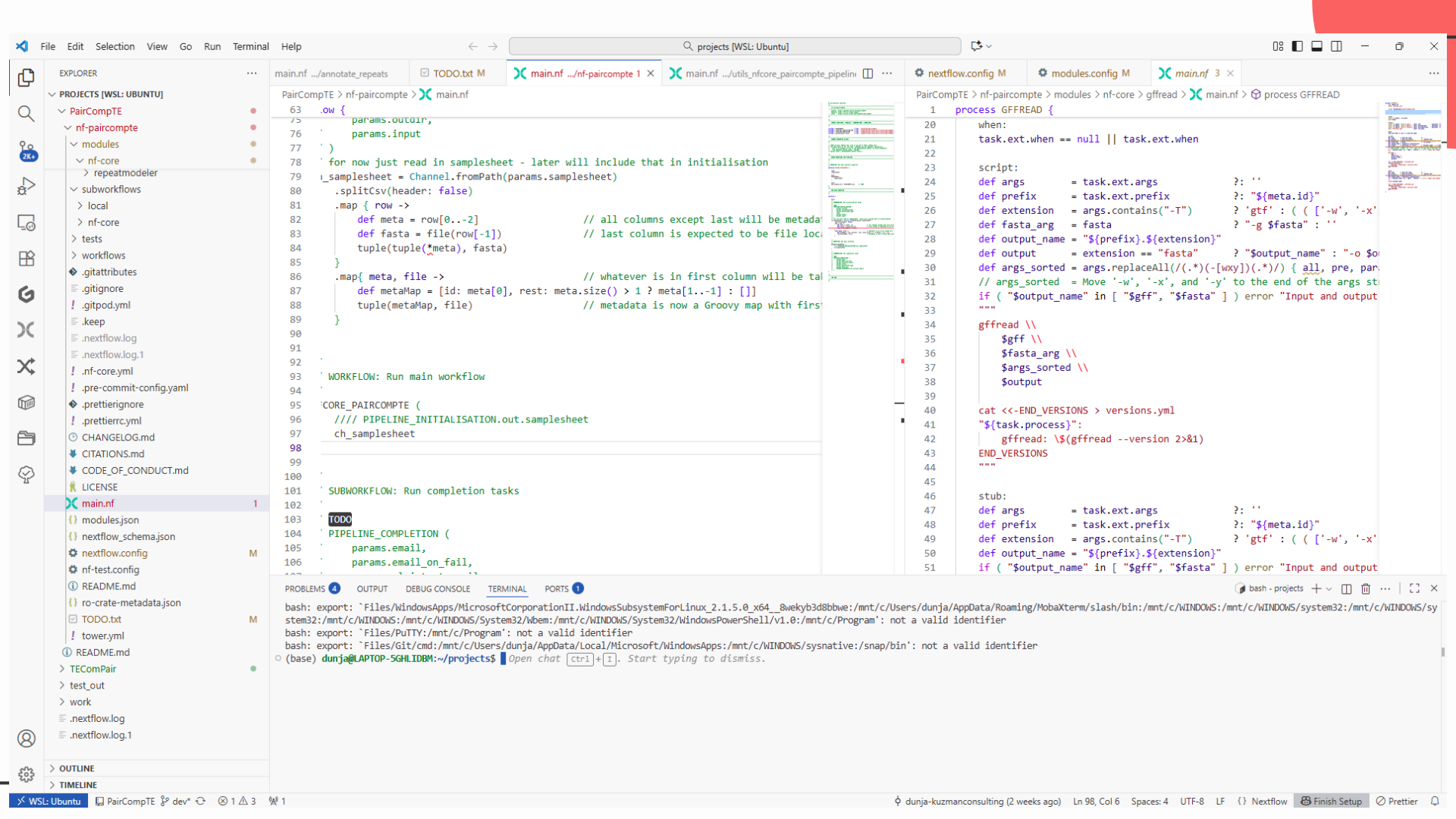
-bash: export: `Files/PuTTY:/mnt/c/Program': not a valid identifier

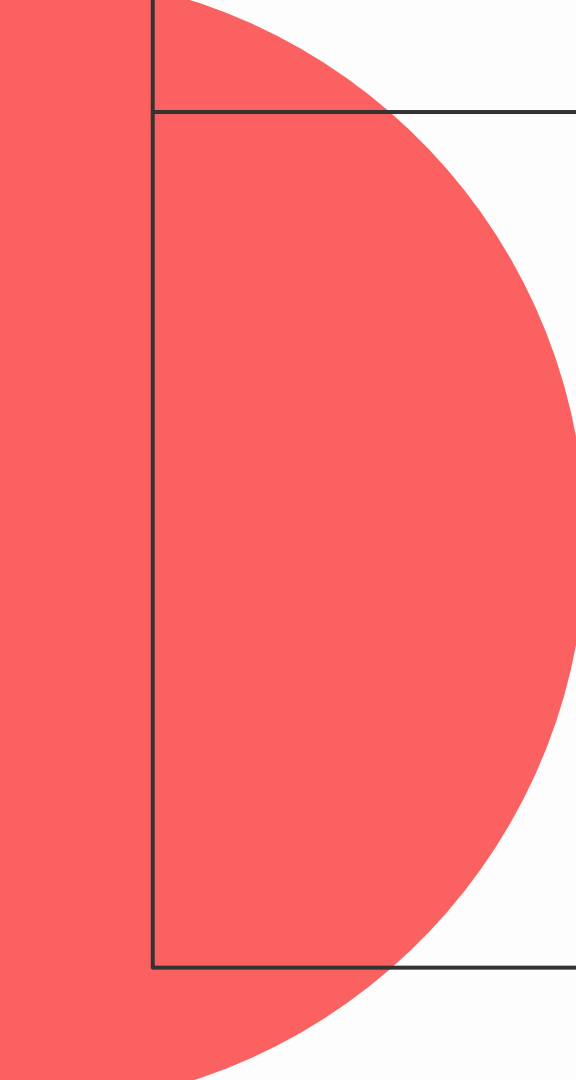
-bash: export: `Files/Git/cmd:/mnt/c/Users/dunja/AppData/Local/Microsoft/WindowsApps:/mnt/c/WINDOWS/sysnative:/snap/bin': not a valid identifier

(base) dunja@LAPTOP-5GHLIDBM:~\$

UNREGISTERED VERSION

- Please support MobaXterm by subscribing to the professional edition here: <https://mobaxterm.mobatek.net>



A large red circle is positioned on the left side of the image, partially cut off by the edge. It overlaps with a white rectangular area that is defined by a thin black border.

Galaxy

Galaxy

EUROPE

- uređeni i održavani skup alata
- “ugrađene” referentne sekvence
- besplatni resursi – uz rizik čekanja

The screenshot displays the Galaxy Europe web interface. On the left is a sidebar with navigation options: Upload, Tools, Workflows, Workflow Invocations, Visualization, Histories, History Multiview, Datasets, Pages, Notifications, and More. The main panel shows an 'Invoked Workflow: GTN Tutorial: M. tuberculosis Variant Analysis (Version: 1)'. The workflow is a complex graph of steps, including '1. BWA-LI', '2. BWA-LI', '3. BWA-LI', '4. BWA-LI', '5. BWA-LI', '6. BWA-LI', '7. BWA-LI', '8. BWA-LI', '9. BWA-LI', '10. BWA-LI', '11. BWA-LI', '12. BWA-LI', '13. BWA-LI', '14. BWA-LI', '15. BWA-LI', '16. BWA-LI', '17. BWA-LI', '18. BWA-LI', '19. BWA-LI', '20. BWA-LI', '21. BWA-LI', '22. BWA-LI'. The workflow is currently running, with a progress bar indicating '22 of 22 steps successfully scheduled' and '16 of 15 jobs complete'. The right sidebar shows the 'History' tab with a search bar and a list of recent datasets, including '81: TB-Profiler Profile on data 66', '82: Results.json', '83: TB Variant Filter on data 66', '79: MultiQC on data 66 and data 64: Webpage', '78: MultiQC on data 66 and data 64: Stats', '77: snippy on data 66, data 59, and data 4 mapped reads (bam)', '76: snippy on data 66, data 59, and data 4 snps table', and '75: snippy on data 66, data 59, and data 4 snps vcf file'.

Welcome to Galaxy, please log in

Public Name or Email Address

Password

Forgot password? Click here to reset your password.

Login

LSLOGIN

IAM4nfdi

EOSC AAI (testing)

Login with EOSC AAI

Don't have an account? [Register here.](#)

korisnički račun - brzo i besplatno!

Galaxy Europe

Tools

Get Data

Send Data

Collection

GENERAL T

Text Mani

Convert F

Filter and

Join, Sub

GENOMIC

Convert F

FASTA/FA

FASTQ Q

Quality C

SAM/BAM

BED

VCF/BCF

Nanopore

COMMON

Operate

Fetch Seq

GENOMICS ANALYSIS

Annotation

Upload

Tools

ChatGXY

Workflows

Workflow Invocations

Interactive Tools

Visualization

Histories

History Multiview

Datasets

Pages

Notifications

More

Galaxy Training!

Contributors

Learning Pathways

Help

Settings

Search Tutorials

History

ets

history

age

story is empty.

d your own data or get

n external source.

Welcome to Galaxy Training!

Collection of tutorials developed and maintained by the worldwide Galaxy community

Galaxy for Scientists

We have separated the tutorials into several categories based on field and technology. We are exploring other ways to organise the tutorials going forward!

Start Here

Topic	Tutorials
Introduction to Galaxy Analyses	15
Using Galaxy and Managing your Data	27

Not sure where to start?

Try the NGS Basics Learning Path!

Start Learning

Scientific Fields

Quickstart

Learning Pathways

Galaxy for SysAdmins

Galaxy for Developers

Galaxy for Teachers

Upcoming Events

Check out upcoming events around the Galaxy!

dokumentacija i

tutoriali

Welcome to the Galaxy Europe Microbiology Lab by the microGalaxy community!

Whether you're working with microbiome samples or bacterial isolates, long or short reads, shotgun or 16S sequencing, genomics, transcriptomics, proteomics, metabolomics, or integrative multi-omics analysis—this is the place for you! Learn more about the Microbiology Lab and the effort of the microGalaxy community in our [preprint](#).

How does this page relate to Galaxy Europe?

Getting started

Data import

Highlight tools

Interactive tools

Learning pathways

Common tools that allow for data import.

Import data to Galaxy

[Download data from NCBI GenBank/RefSeq](#)

[Download raw reads from NCBI SRA](#)

[Download run data from EBI SRA](#)

- Genome Assembly
- Nanopore
- FinchusMC
- Climate
- HICE Explorer
- Genome Annotation
- Metabolomics
- microGalaxy
- Biodiversity Genomics
- RNA
- Ecology
- Proteomics
- Cheminformatics
- Imaging
- StreetScienceCommunity
- Single Cell Omics
- Human Cell Atlas
- Machine Learning
- Plants
- Virology
- Lite
- Phase
- Earth System
- Astronomy

ry

+

≡

datasets

▼

✕

ned history

✎

ilt Storage

📍

🔄

is history is empty.

an load your own data or get
rom an external source.

specijalizirane pod-stranice olakšavaju izbor alata

Galaxy Europe

Using 2% of 250.0 GB

dunjag

ChatGXY

Workflows

Workflow Invocations

Interactive Tools

Visualization

Histories

History

Datasets

Libraries

Recent Exports

Notifications

More

Search

exclude restricted

Name	Description	Synopsis	
covid-19	Public data related to COVID-19	RAW sequence data & VCFs	
Critical Assessment of Metagenome Interpretation (CAMI)	CAMI datasets		
Earth System Community Modeling	Input data used for running simulation w ... (more)	Earth System Community Modeling data	
Galaxy courses	Data for Galaxy courses		
Genomes + annotations	Reference genomes and gene annotations		
GTN - Material	Galaxy Training Network Material	Galaxy Training Network Material. See ht ... (more)	
IAEA workshop	workshop data		
Mira_test	for testing		
PGP-UK Open Access Data	Open Access Genomic, Transcriptomic and ... (more)	https://www.personalgenomes.org.uk	
Street Science Community	Data for the Street Science Community		

«

◀

1

▶

»

10

per page, 10 total

> History

search datasets

Unnamed history

Default Storage

This history is empty.

You can load your own data or get data from an external source.

javno dostupni podatci

Galaxy Europe

Using 2% of 250.0 GB | dunjag

Upload

Tools

ChatGXY

Workflows

Workflow Invocations

Interactive Tools

Visualization

Histories

History Multiview

Datasets

Pages

Search

exclude restricted

Name

covid-19

Critical Assessment of Metagenome Interpretation (CAMI)

Earth System Community Modeling

Galaxy courses

Genomes + annotations

GTN - Material

IAEA workshop

Mira_test

PGP-UK Open Access Data

Street Science Community

Upload from Disk or Web to Unnamed history

Regular Composite Collection Rule-based

Drop files here

Type (set all): Auto-detect Reference (set all): unspecified (?)

Choose local file Choose from repository Paste/Fetch data Start Pause Reset Close

« < 1 > » 10 per page, 10 total

History

search datasets

Unnamed history

Default Storage

0 B

This history is empty. You can load your own data or get data from an external source.

vaši podatci

Galaxy Europe

Tools

Discover Tools

search tools

Discover Tools in this Galaxy

search tools

Select a section to filter by Select an ontology to filter by

Ontologies Favorites

Display:

History

search datasets

Unnamed history

Default Storage

9 B

This history is empty. You can load your own data or get data from an external source.

Get Data

Collection Operations

GENERAL TEXT TOOLS

Text Manipulation

Convert Formats

Filter and Sort

Join, Subtract and Group

GENOMIC FILE MANIPULATION

Convert Formats

FASTA/FASTQ

FASTQ Quality Control

Quality Control

SAM/BAM

BED

VCF/BCF

Nanopore

COMMON GENOMICS TOOLS

Operate on Genomic Intervals

Fetch Sequences / Alignments

GENOMICS ANALYSIS

Annotation

DICED Database DICED is a web interface for access to proteolytic peptides

Get Data

Copy Link Favorite Tool Open

Upload File from your computer

Show tool help

Get Data Query and retrieval

Favorite Tool Open

SEEK test Trying to get open files out of SEEK

Get Data

Copy Link Favorite Tool Open

UCSC Main table browser

Get Data Query and retrieval

Copy Link Favorite Tool Open

UCSC Archaea table browser

Get Data Query and retrieval

Copy Link Favorite Tool Open

NCBI Datasets Genomes import data from the NCBI Datasets Genomes page

Get Data Query and retrieval

Copy Link Favorite Tool Open

downloads via lftp

This tool downloads a list of URLs via lftp and creates a collection of files.

dobro
održavani
alati

Galaxy Europe

Using 2% of 250.0 GB | dunjag

Select Histories

search histories

Select histories to pin to Multiview

- ☐ Unnamed history (0 items)
about 3 hours ago
- ☐ RNAseq (9 items)
about 15 hours ago

- All histories loaded -

History Multiview

Search datasets and collections in selected histories

Recent Select

Current History Switch to

Unnamed history

Last edited about 3 hours ago

Default Storage

0 B

☒ This history is empty.
You can load your own data or get data from an external source.

RNAseq

Last edited about 15 hours ago

Default Storage

5.86 GB

- ☒ 3: Single-end data (fastq-dump)
a list with 6 fastqsanger.gz datasets
- ☒ 2: Paired-end data (fastq-dump)
a list with 0 pairs
- ☒ 1: SRR_Acc_List.txt

+ Create and pin new history

Select histories

History

search datasets

Unnamed history

Default Storage

0 B

This history is empty.
You can load your own data or get data from an external source.

History = povijest provedenih analiza

- moguće više analiza paralelno
- dijeljenje
- workflows